

# Praveena Kaliyaperumal

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## PROFESSIONAL SUMMARY

Software Engineer with 7+ years of experience in building and deploying scalable applications using Python and SQL, following SDLC, Agile, and Waterfall methodologies. Expert in backend development with Django and Flask, designing RESTful APIs, and ensuring seamless system integration. Proficiency in front-end development using React and Angular, implementing responsive web interfaces with HTML, CSS, JavaScript, Bootstrap, Redux, and AJAX. Well-versed in leveraging Python libraries such as Pandas, Scikit-learn, TensorFlow, Seaborn, SciPy, Matplotlib, for data analysis, visualization, and machine learning applications. Understanding of cloud technologies like AWS, Azure, and DataBricks for deploying and managing cloud-native applications. Hands-on experience with Docker for containerization, improving scalability and deployment efficiency. Well-versed in working with relational and NoSQL databases, including MySQL, PostgreSQL, SQL Server, Firebase, and MongoDB, optimizing queries and ensuring data integrity.

## SKILLS

**LLM & AI Tools:** Gemini, Claude, Anthropic, OpenAI, Vertex AI, RAG pipelines

**Programming Languages:** Python, SQL, PL/SQL, JavaScript, Typescript, Kotlin

**Packages:** NumPy, Pandas, Matplotlib, TensorFlow, Scikit-learn, Seaborn, SciPy

**Prototyping & Visualization:** Figma, InVision, Balsamiq, Wireframing, Storyboarding

**Frameworks:** Django, Flask, Node.js, Express.js, React, Angular

**Cloud Technologies:** AWS, GCP, Azure, Databricks, Docker, Snowflake, Airflow (Cloud Composer), CI/CD

**Web Technologies:** HTML, CSS, JavaScript, AJAX, XML, JSON, jQuery, Bootstrap, Restful API, Redux, Kotlin

**Databases:** MySQL, SQL Server, MongoDB, Firebase, PostgreSQL

**Version Control/Other Tools:** GitHub, Git, Jira

**Operating Systems:** Windows, Linux, Mac

## WORK EXPERIENCE

### Full Stack Engineer | ChekOut.AI, USA

July 2025 - Present

- Designed and deployed AI-powered chatbot pipelines using Google Cloud Functions, Vertex AI, and OpenAI/Gemini/Claude models for multiple brands.
- Implemented Retrieval-Augmented Generation (RAG) with Cloud Storage, Big Query, and Looker, enabling end-to-end analytics and business insights.
- Built backend APIs, logging infrastructure, and automated lead scoring pipelines in Big Query/Looker Studio to evaluate conversation quality and user engagement.
- Developed Airflow (Cloud Composer) DAGs for scheduled data processing and analytics reporting across chatbot instances.
- Resolved latency and security issues in Cloud Functions and Big Query ingestion, ensuring scalable and reliable system performance.
- Collaborated with cross-functional teams on Shopify integration and weekly product demos, aligning technical features.

### Software Engineer | CoxPHIT, USA

Mar 2023 – July 2025

- Considered and implemented user-friendly front-end interfaces using React and Angular, improving UI responsiveness and user engagement by 25%, with seamless functionality across desktop and mobile platforms.
- Identified software failures and data issues, providing metrics reports to management for informed decision-making.
- Automated internal audit procedures for Medicaid and HIPAA compliance while conducting ongoing field research to stay update on regulatory standards.
- Calculated and maintained RESTful APIs using SQL and Flask, ensuring efficient data processing and reducing API response time.
- Streamline operational framework related to medical billing, payroll, and reconciliation, using Selenium and Python which have increased accuracy and eliminated 90% of manual work.
- Enhanced PostgreSQL databases, abstracting and managing complex schemas, writing efficient queries, and improving database performance by 40%, ensuring consistent data integrity across systems.

### Software Engineer | Accenture, India | Client - Bank of America

Jul 2019 - Dec 2022

- Designed, developed, and organized scalable applications using Agile methodologies, leveraging Python, Django, and PL/SQL to meet project requirements and ensure timely delivery of high-quality software.
- Conceptualized Python libraries like NumPy, Pandas, and Matplotlib to analyze datasets of over 1 million records, creating data visualizations that supported key business decisions and improved reporting accuracy by 30%.
- Collaborated with cross-functional teams to track transaction monitoring compliance and identify areas for process improvement. Worked extensively with Google Cloud Platform (GCP) and AWS for deployment and cloud-based testing environments.
- Established Python scripts for automated testing of workflows in a financial services project, ensuring alignment with project requirements. Observed service delivery concerns, participating in planned and ad hoc client meetings to resolve issues effectively.

- Delivered software projects using the Waterfall methodology, ensuring clear documentation and timely completion.
- Industrialized and structured web applications using Flask, Node.js, and Express.js, enhancing system performance and optimizing response times for high-traffic environments.
- Developed ML models with TensorFlow, Seaborn, etc., to extract insights from large datasets and support data-driven decisions.
- Optimized database queries in MySQL and SQL Server, reducing query execution time by 40% and enhancing data consistency and integrity across enterprise applications, supporting high-volume transactions.
- Planned front-end features using HTML, CSS, JavaScript, and jQuery to enhance user experience, interactivity, and responsiveness.
- Deployed scalable applications on AWS, ensuring high availability, system reliability, and optimized structure cost management.

EDUCATION

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|-----------------------------------------------------------------------------|-----------------|
| <b>Master of Science in Data Science</b>                                    | <b>Feb 2025</b> |
| Saint Peter’s University                                                    | New Jersey      |
| <b>Bachelor of Engineering in Electronics and Communication Engineering</b> | <b>Apr 2019</b> |
| Meenakshi Sundararajan Engineering College                                  | Chennai, India  |